

Optimized Automated Suture Needle Segmentation and Re-grasping Using the Da Vinci System

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Abstract—This paper utilizes the Da Vinci Research Kit (dVRK) for automated circular needle picking and regrasping tasks, presenting an optimized YOLO (You Only Look Once) model for suture needle segmentation and a re-grasping algorithm for suturing. The paper describes the process of autonomous suture needle grasping, manipulated and regrasped between different robot arms. We succeed grasping over 38 over a total of 40 needles with diverse backgrounds and conditions. Failures occurred due to the segmentation and motion planning errors.

I. INTRODUCTION

Suturing is a demanding task in modern surgeries, requiring manual identification of critical components such as target points and controlling robot movement. Automated suturing can reduce surgeon fatigue, operating time, and enable supervised tele-surgery [1]. Target detection, tracking, and grasping are time-consuming tasks, particularly in manual suturing. The Da Vinci Research Kit (dVRK) has enabled automated target detection and grasping during suturing. This paper proposes an optimized learning model trained with a dataset for needle tracking and manipulation using the Da Vinci surgical robot arm.

II. METHOD

A. Circular Needle Segmentation

The You Only Look Once (YOLO) algorithm version 8 is a CNN-based object detection framework that has shown significant improvements in both accuracy and speed [2] with respect to its predecessors, making it one of the most efficient and effective models for real-time object detection tasks. It was used in our work for segmenting needle within an image from the endoscope of da Vinci system and generate pixel-wise masks for each circular needle found in an image. an image.

B. Model Optimization

We optimized our system by diversifying the dataset and fine-tuning hyperparameters. We employed data augmentation techniques to create realistic views of needles, simulating those commonly encountered in surgical settings and operations. Specifically, we applied various image transformations, both linear and non-linear, such as flipping, rotating,

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cropping, grayscale conversion, hue adjustment, brightness adjustment, exposure adjustment, blurring, and adding noise. The dataset annotations cover complex areas and feature diverse needle quantities and backgrounds to enhance the model's robustness.

Hyperparameter tuning iteratively refined loss functions and learning rates to optimize model accuracy. Increasing iterations typically reduce segmentation loss during training, though excessive iterations may lead to overfitting indicated by increased validation segmentation loss.

C. Needle Grasping

The planning of the tool tip trajectory involves moving in such a way that the gripper grasps the needle at an angle ϕ degrees above the end point of the needle with the thread. The orientation of the gripper will be along the tangent angle of the pickup point. The needle will be positioned perpendicular to the platform surface and rotated by ϕ degrees.

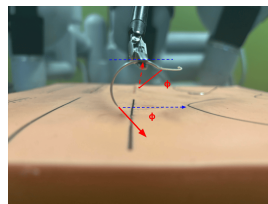


Fig. 1: An example of grasping with the needle's pickup point and the setup for the suture.

III. EXPERIMENT

A. Validation of the pre-trained YOLO v8 model

The machine learning model used for the needle's suture segmentation was validated to assure precise manipulation. The test dataset was used to predict the pixel coordinates of the needle tip, and the opposite end. The location detection error is used to assess the model's performance. We did the grasping test of test cases on the real system. Euler Distance was used to assess performance between the predicted end point pixel and the ground truth pose position. The distance was measured as:

- $mask_x, mask_y$: (x, y) pixel coordinates of the needle end point predicted from the YOLO model.
- g_x, g_y : ground truth suture needle end point pose.
- D_{fitted} : the diameter of the fitted circle of the circular suture needle.

$$\text{Distance ratio} = \frac{\sqrt{(\text{mask}_x - g_x)^2 + (\text{mask}_y - g_y)^2}}{D_{\text{fitted}}} \quad (1)$$

The Least Squares Method (LSM) was used to find a fitted circle around the contour of the mask and the pickup point of the suture needle. The angle ϕ is defined as 30 degrees in the experiments.

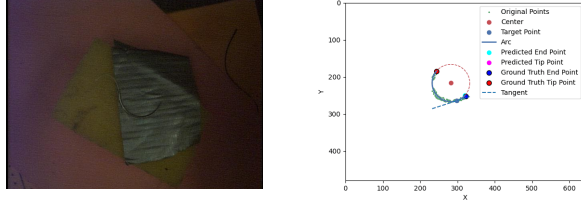


Fig. 2: An example of test data (A needle set on a flat platform) and data processing outputs

The outputs after the segmentation and localization are showed in Figure 2:

- Target point, Tangent: The desired pickup point on the needle and the desired orientation of the end-effector while grasping.
- Arc: The predicted body of suture needle.
- Predicted End Point, Predicted Tip Point: The pixel coordinates of the needle tip, and the opposite end of the suture needle.
- Ground Truth End Point, Ground Truth Tip Point: The ground truth pixel coordinates of the two needle tips of the suture needle.

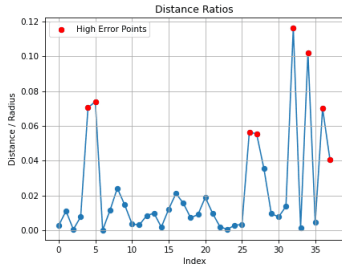


Fig. 3: The distance ratio from the tests on 38 test data and the high error data.

To verify the performance of the grasping point detection algorithm, the Da Vinci surgical system was used for testing the grasping algorithm. The goal was to determine whether those errors would lead to failures in grasping. Successful grasping and manipulation of the suture needle along a specified trajectory indicates acceptable performance. These assessments provide critical error metrics that directly validate the algorithm's operational effectiveness. Figure 3 shows the distance ration(err) of the test data and the data for the real system grasping tests with high error output.

B. Suture needle grasping and regrasping

Grasping and regrasping the needle are crucial steps in suturing. The Da Vinci robotic system uses image data from cameras on its endoscope. The pixel coordinates of the needle pickup point are used by the robot arm (PSM_1) to grasp the needle and carry it along a suturing trajectory. After the initial suture, the needle is regrasped by the second robot arm (PSM_2).

The robot arms are positioned using Cartesian coordinates, while the needle's pickup point is determined by pixel-wise coordinates from the endoscope. The matrix H transforms these pixel coordinates $[u, v]$ into the x, y, z position in the camera frame [3]. The homogeneous transformation matrix from the camera frame (c) to the world frame (w) is:

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} = H \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \quad (2)$$

$$ECM_B T_W = ECM_B T_{PSM_B} \cdot PSM_B T_{PSM_T} \cdot PSM_T T_W \quad (3)$$

$${}^c T_W = {}^c T_{ECM_T} \cdot ECM_T T_{ECM_B} \cdot ECM_B T_W \quad (4)$$

- ECM_T and ECM_B : The coordinate frame of the tool-tip and the base of the ECM arm.
- PSM_3T and PSM_3B : The coordinate frame of the tool-tip and the base of PSM.
- W: W is an arbitrarily selected world coordinate system used during the registration process.

The grasping and regrasping test shows in Figure 4.

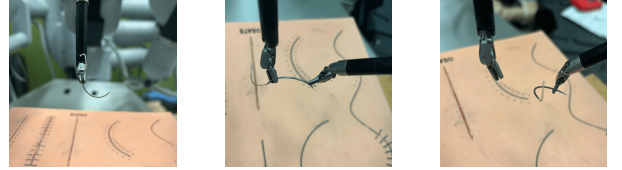


Fig. 4: The grasp, manipulation and regrasp process by using the PSM_1 and PSM_2 .

IV. CONCLUSIONS

This paper introduces a system for autonomous suturing with the dVRK, including methods for needle detection, segmentation, and grasping. Grasp and regrasp operations are pivotal for autonomous suturing. Future work includes optimizing the motion planning algorithm for the entire suture task and comparing it with human performance in terms of precision, errors, and completion time.

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